

PROJECT AND TEAM INFORMATION

Project Title

EV-OPT: Intelligent EV Charging Station Recommendation and Smart Charging Management

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

Nowadays, more and more people are using electric vehicles (EVs). Because of this, EV charging stations can become very crowded. There may be long waiting lines, fewer available chargers, and too much electricity being used at the same time.

Usually, people choose a charging station based only on distance. But the nearest station may be crowded, expensive, or have slow chargers. So, choosing the best charging station can be difficult.

EV-OPT is a smart system that helps EV drivers choose the best charging station. It checks different things such as:
* How far the station is * How many EVs are waiting * How many chargers are available * How fast the charger can charge * The charging cost * How much power the station can provide * How urgently the EV needs charging

The system compares these factors and suggests the most suitable charging station to the driver.

After a driver reserves a charger, EV-OPT also helps decide **which EV should be charged first and how much power each EV should get**. It makes sure that the charging station does not use more power than it can safely provide.

The main goal of EV-OPT is to **reduce waiting time, use charging stations properly, and avoid too much load on the electricity system**.

This project also shows how **C data structures and C++ OOP concepts** can be used together to solve a real-life problem.

In simple words, **EV-OPT is a smart system that helps EV drivers find and use charging stations in a better and more organized way.**

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

Today, EV charging apps mainly help people find charging stations. They show information like the station location, distance, charger type, whether a charger is available, and sometimes the charging cost and charging time.

Most people choose a charging station because it is **nearby or currently available**. But this may not always be the best choice.

For example, a nearby station may have many EVs waiting in a long line. Another station that is a little farther away may have more free chargers and less waiting time. Also, when many EVs need to charge at the same station, it can become difficult to decide which EV should get power first and how much power each EV should receive.

Researchers have already worked on topics like finding routes for EVs, selecting charging stations, planning charging times, and reducing pressure on the electricity grid. However, many of these solutions use complicated mathematical methods.

EV-OPT aims to provide a simpler solution. It considers different factors such as:

* Travel distance and route * Number of EVs at the station * Available chargers * Charging cost * How urgently an EV needs charging * Charging speed * How much power the station can provide

Using these factors, EV-OPT will **suggest the most suitable charging station** and help manage the charging of EVs in a smart way.

The system will also be compared with the simple **nearest available station method**. This comparison will help us check whether EV-OPT can reduce waiting time, use charging stations better, and improve the overall charging process.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

The main goal of **EV-OPT** is to create a smart system that helps EV drivers find the best charging station according to the current situation.

The system will check different things before suggesting a station, such as:

* Travel time * Waiting time * Number of free chargers * How busy the station is * Charging cost * How urgently the EV needs charging

EV-OPT will also allow users to **reserve a charger**, decide which EV should be charged first, and distribute charging power. It will make sure that the station and electricity grid do not receive more load than they can handle.

This project will also show how **Data Structures in C** and **Object-Oriented Programming (OOP) in C++** can be used together in a real-life smart-mobility project.

EV-OPT will have separate screens for **EV drivers and charging-station operators**. It will also provide simple reports and information to show how well the system is working.

The main aim is to help **drivers get a charger with less waiting** and help **station operators manage their charging stations better**.

Planned Milestones

1. Requirement Analysis Understand the needs of EV drivers and station operators. Decide what information the system will take, what it will give as output, and what rules it must follow.

2. System Design Plan the complete system, including its modules, data flow, database, and user interface.

3. Data Structure Implementation Use important data structures and algorithms in C, such as **Graph, Dijkstra's Algorithm, Hashing, Queue, Priority Queue, Sorting and Searching**.

4. C++ OOP Development Create different parts of the system using C++ OOP, such as **EV, Station, Charger, Charging Request, Reservation, Session, Scheduler and Recommendation**.

5. Core Intelligence Develop the main logic for **station recommendation, charging priority, scheduling and power allocation**.

6. UI and Integration Connect the user interfaces for EV drivers and station operators with the main system.

7. Testing and Evaluation Test the system using different situations and compare EV-OPT with the simple **nearest available station** method.

8. Finalization Study the results, prepare the project documentation and research paper, and give the final project demonstration.

Planned milestones:

- **Requirement Analysis** – Identify stakeholders, inputs, outputs, constraints and functional requirements for the EV-OPT system.
- **System Design** – Finalize architecture, modules, data flow, database structure and UI workflow for the EV-OPT system.
- **Data Structure Implementation** – Implement Graph, Dijkstra, Hashing, Queue, Priority Queue and relevant sorting/searching techniques in C for the EV-OPT system.
- **C++ OOP Development** – Develop EV, Station, Charger, Charging Request, Reservation, Session, Scheduler and Recommendation components for the EV-OPT system using object-oriented C++.
- **Core Intelligence** – Implement station recommendation, charging priority, scheduling and power-allocation logic for the EV-OPT system.
- **UI & Integration** – Integrate EV-user and station-operator interfaces with the complete EV-OPT system.
- **Testing & Evaluation** – Conduct simulated scenarios and compare the EV-OPT system with a nearest-available-station baseline.
- **Finalization** – Analyze results, prepare documentation, research paper and final demonstration for the EV-OPT system.

PROPOSAL DESCRIPTION (CONTINUED)

Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

I will develop **EV-OPT** as a smart software system that works in five simple steps:

Input → **Processing** → **Intelligence** → **Action** → **Feedback**

First, the **EV user** will enter information such as current location, battery level, required battery level, destination and the time by which charging is needed.

The **station operator** will provide station information such as available chargers, number of EVs waiting, power capacity, current power usage, charging price and maintenance status.

The system will use different **Data Structures and Algorithms** to process this information. A **Graph** and **Dijkstra's Algorithm** will be used to find routes and calculate travel distance or time. **Hashing** will help the system quickly find EV and station information. **Queue and Priority Queue** will be used to manage charging requests and decide which EV should be charged first. Sorting and searching techniques will help rank and find suitable charging stations.

The main application will be developed in **C++** using OOP concepts such as:

* **Classes** * **Encapsulation** * **Inheritance** * **Polymorphism** * **Abstraction**

The main parts of the system will include **EV, User, Station, Charger, Charging Request, Reservation, Charging Session, Scheduler and Recommendation Engine**.

The Data Structures written in **C++** will be connected with the C++ application. The **Recommendation Engine** will compare different charging stations using factors such as travel time, waiting time, available chargers, station load, charging cost and how urgently the EV needs charging.

After a user reserves a charger, the **Scheduler** will decide the charging order and distribute power to the EVs while making sure that the station and power grid are not overloaded.

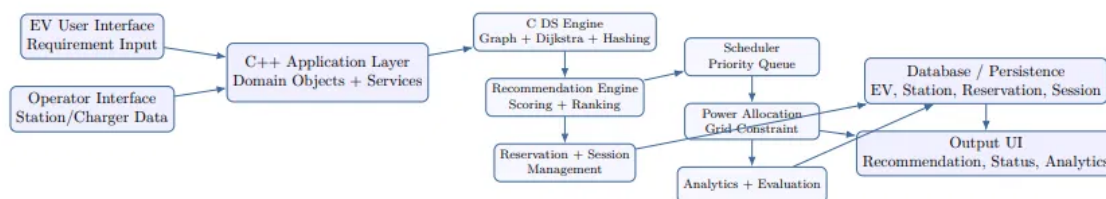
A **web-based interface** will be created using **HTML, CSS and JavaScript**. It will have separate screens for EV users and station operators.

MySQL will be used to store the system's data, while **Git and GitHub** will be used to save different versions of the project and support teamwork.

Finally, the system will be tested using different charging situations. Its results will be compared with the simple **nearest available station** method to check how much EV-OPT improves the charging process.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



End-to-end workflow: EV requirement + station condition → route/candidate analysis → multi-factor recommendation → reservation → priority scheduling → constrained power allocation → charging-session simulation → persistence and analytics.

PROPOSAL DESCRIPTION (CONTINUED)

Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

The primary deliverable will be a working EV-OPT software prototype demonstrating intelligent EV charging-station selection and charging management. Expected deliverables include:

- C-based graph, Dijkstra, hashing, queue and priority-queue components.
- C++ OOP application layer representing EVs, stations, chargers, requests and sessions.
- Multi-factor station recommendation and ranking engine.
- Reservation and priority-aware charging scheduler.
- Constraint-aware power-allocation simulation.
- User and station-operator interfaces with charging-session status.
- Persistent records and analytics for waiting time, cost, utilization, load and deadline satisfaction.
- Comparative evaluation against a nearest-available-station baseline.

The architecture is intended to remain extensible for future integration with real charging hardware or external station APIs.

Assumptions (1 pt)

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words).

The system will use a **road map** and charging stations with proper information about the stations and chargers.

The station operator will keep the information updated. This information will include whether a station is open, how many EVs are waiting, which chargers are free or busy, how much power the station can give, how much power it is using and the charging cost.

The charging process will be done **inside the software**. The system will not control real charging machines. The travel time and charging time will be calculated using the available information.

The amount of power available from the electricity grid can also be changed when needed.

The system will be tested in different situations, such as:

* Normal charging * Busy charging times * Traffic jams * Chargers not working * Less power available from the grid

These tests will help check if **EV-OPT** can work properly in different situations.

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

1. International Energy Agency (IEA). (2026). *Global EV Outlook 2026: Electric Vehicle Charging*. International Energy Agency.
2. International Energy Agency (IEA). (2025). *Global EV Outlook 2025*. International Energy Agency.
3. Dijkstra, E. W. (1959). "A Note on Two Problems in Connexion with Graphs." *Numerische Mathematik*, 1, 269–271.
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5. Open Charge Alliance. *Open Charge Point Protocol (OCPP)*. Open Charge Alliance.
6. ISO/IEC. (2024). *ISO/IEC 14882:2024 – Programming Languages – C++*. International Organization for Standardization.